

Method Report — Billing Event Reconstruction and Payment-Delay Early-Warning for Clinical Laboratory Settings

- Version: 2.0
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1. Research Questions

- RQ1: Under multi-source, missing or conflicting records, does uncertainty-preserving event reconstruction (explicit lineages, conflicts, needs-review queue) reduce wrong links and control manual review volume, compared with simple identifier joins and appropriate existing methods?
- RQ2: Does incorporating available record-quality and processing-stage signals improve within-horizon prediction of first-observable positive payment allocation vs. no-payment closure vs. still-open, relative to aging-rule and basic baselines?

2. Related Work and Position

- Fellegi & Sunter (1969): decision-theoretic record linkage with a possible-link disposition; the needs-review tier operationalizes this disposition.
- Christen (2012) and van der Aalst (2016): general matching and event-log analysis; this work adds billing-domain event semantics and as-of point-in-time discipline.
- Deep Claim (arXiv:2007.06229): predicts payer responses pre-submission from claims; this work differs in scope: post-submission billing-event reconstruction plus an observation-window-aware competing-event forecast.
- Fine & Gray (1999) / Kaplan & Meier (1958): competing-risk and censoring foundations; a discrete-time interval formulation is adopted for irregular observation.

- Zeng et al. (2007), Schoonbee (2022): invoice-lateness prediction; the aging baseline is informed by these, with explicit censoring treatment added.

Reference list: research/prior_work.csv (nine references). Reading scope is recorded per reference: bibliography-verified (metadata confirmed against the publisher's record) or bib-and-abstract-verified (metadata plus abstract read in full). Full-text analysis is not claimed for any reference at this stage.

3. Data and Permissions

- The study uses de-identified billing event records from three participating clinical laboratory organizations, referred to by site tokens CPL, TRI and NDX under the data-use arrangement. Site tokens are used throughout; organizational identities are recorded in the study documentation.
- Data window: 2022-07-01 to 2023-11-15. Records carry per-site local timezones (including daylight-saving transitions); submission and payment activity follows weekday-dominant rhythms with a mix of same-day and lagged record entry.
- Data quality issues observed and handled: duplicate imports, cross-organization duplicate bill numbers, missing correction references, timezone-less timestamps, invalid amounts and non-USD currencies. These are isolated by the schema validator, never silently repaired, and reported in the run documentation.
- Manually adjudicated reference labels (record pair lineage membership and observed outcome times) are held out from all model inputs and used only for evaluation.
- Real-mode processing of identified data remains disabled; see governance/data_permissions.json.

4. Event Dictionary and Prediction Target

- Event types: SUBMITTED, CORRECTED, PAYMENT_POSTED, PAYMENT_REVERSED, VOID_VERSION,

CLOSED_NO_PAYMENT, REOPENED, ADJUSTMENT, STATUS_UPDATED, UNKNOWN (with review).

- Target: within $H=30$ days after landmark t (first observable submission + 7 days), predict the first observable positive payment allocation, first no-payment closure, or still-open. Not final bank cash, not full settlement.
- Protocol v1.1 (PC01, 2024-08-06): interval granularity 1 -> 5 days after sparsity findings at 1-day resolution.

5. Candidate Methods

- Module A: A0 exact-id baseline (native previous_record_id with org/cycle/time checks); A1 constrained linkage (R2 same-bill-id version chains; R3 amount/date-window weak evidence with uniqueness gating). Accepted / rejected / needs-review tiers; no unconstrained transitive chaining; manual decisions applied as a separate layer.
- Module B: B0 age/stage empirical-risk baseline (abstains on unsupported cells); B1 discrete-time multinomial logistic regression; C1 B1 plus reconstruction-quality features. Class probabilities per interval with the product-rule cumulative curves; sum-to-one enforced (T16).

6. Splits and Leakage Control

- Temporal split grouped by lineage, 60/20/20; test portion locked.
- Features computed strictly from records observable at t (T10); reference labels and outcome fields excluded from features (T13); lineages never span splits (T14); preprocessing fitted on train only (T15).

7. Results (locked test set)

Cohort flow (all records of the reference run):

bills 1,164 -> reconstructed lineages 1,044 -> snapshots 959

-> locked test snapshots 192. Snapshots whose outcome was already

determined before the landmark are not prediction targets and are excluded at snapshot construction.

Module A (held-out adjudicated adjacency evaluation):

- reference pairs 95; predicted 123
- true positives 94; false positives 29; false negatives 1
- precision 0.7642 (95% Wilson interval 0.680-0.832); recall 0.9895 (0.943-0.999); refusal ratio 0.0956
- AO unconstrained baseline (same-org chaining in submission order, no evidence requirements): precision 0.0017, recall 0.0211. The constrained method's precision is materially higher (0.7642 vs 0.0017); the AO baseline is retained only as a deliberately weak lower bound and is not presented as a competing method.
- The weak-evidence rule R3 is the dominant false-positive source; multi-candidate ambiguity is routed to review instead of being forced.

Module B (snapshot-level, n=192 test snapshots; outcome mix

{"payment": 576, "still_open": 375, "close_no_payment": 8} over all 959 snapshots):

- B0: snap log-loss 0.7149, Brier 0.4943
- B1: snap log-loss 0.7123, Brier 0.4908 (absolute delta -0.0026; relative improvement 0.36%)
- C1: snap log-loss 0.7124, Brier 0.4915 (delta vs B1 +0.0001 — no incremental value)
- The B1-vs-B0 delta of 0.36% does not clear any practically meaningful improvement threshold; it is reported as a neutral result. No pre-specified gain margin was frozen in the protocol; that omission is

recorded as a study-design limitation (section 9).

8. Negative / Neutral Findings (reported as-is)

- C1 quality features add no measurable value on the locked test set.
- An early implementation with balanced class weighting distorted conditional probabilities approximately 4x; corrected after diagnostics (2024-12).
- Close-no-payment events are rare in the observation window (n=8); the model abstains rather than fabricating probabilities where support is insufficient.
- The candidate B1 improves snapshot log-loss over the B0 baseline by only 0.36% on the locked test set — not a substantive gain.

9. Applicability and Limitations

- Applicable condition: multi-source billing exports with availability-time semantics, USD-denominated amounts, explicit or inferable version chains.
- Not applicable: final collection forecasting, cross-currency aggregation, identified patient-level analytics without authorization.
- Limitations: the participating sites span three ownership models in one national context; generalization to other settings requires further validation. Landmark and horizon are fixed study defaults pending operational confirmation. No external independent validation has been completed to date. No pre-specified gain margin for the Module B comparison was frozen in the research protocol; the post-hoc interpretation of the observed 0.36% delta is therefore descriptive, not confirmatory. Uncertainty intervals for Module B snapshot-level differences are not reported because the snapshot estimates are not independent.

10. Contributions

- The research design, decision records (research/decisions/, D01-D06 and PC01) and

the analysis were carried out by the applicant; engineering execution used computational tooling under her direction. Individual decisions trace to code and tests as documented in the decision records.

11. Reproducibility

- Environment: environment.json (Python 3.12.13; historical snapshot 3.11.5).
- Commands: the standard pipeline entry (full pipeline); pytest (32 acceptance and regression tests); the validation entry (held-out evaluation). Full command references in README.
- Code and data access: <https://github.com/fanshiyu35/lab-billing-research>
- Run referenced: the v1.4 reference run (artifacts archived with the code release).

References

See research/prior_work.csv for the verified reference list.